

Lab A0 – Sim

Introduction to Electronic Simulation

This lab introduces electronic simulation using LTspice. LTspice is an electronic circuit simulation based on the open-source simulation engine SPICE (Simulation Program with Integrated Circuit Emphasis). In the future, you may run into other variants of SPICE such as HSPICE, PSPICE, etc. but they all use a similar underlying algorithm. Follow [this link](#) to get more information about the SPICE algorithm (this material isn't tested in this course but is useful to understand).

The LTspice software is free, and it is available for all types of OS: Windows, MacOS and Linux. However, the MacOS version is significantly different from the Windows/Linux version. **We advise you to use the lab computers for any work done in the labs, as the instructions for these labs are for Windows/Linux.** If you decide to work outside the lab and have a Mac, please be aware of the differences; the capability is the same but many options are hidden in harder to find menus. TAs should always be able to help you independently of your OS, and online tutorials are also always your friend!

Today, there are three aspects of simulation that we will be look at – operating point simulation (.op), time domain transient analysis (.tran), and frequency domain AC analysis (.ac).

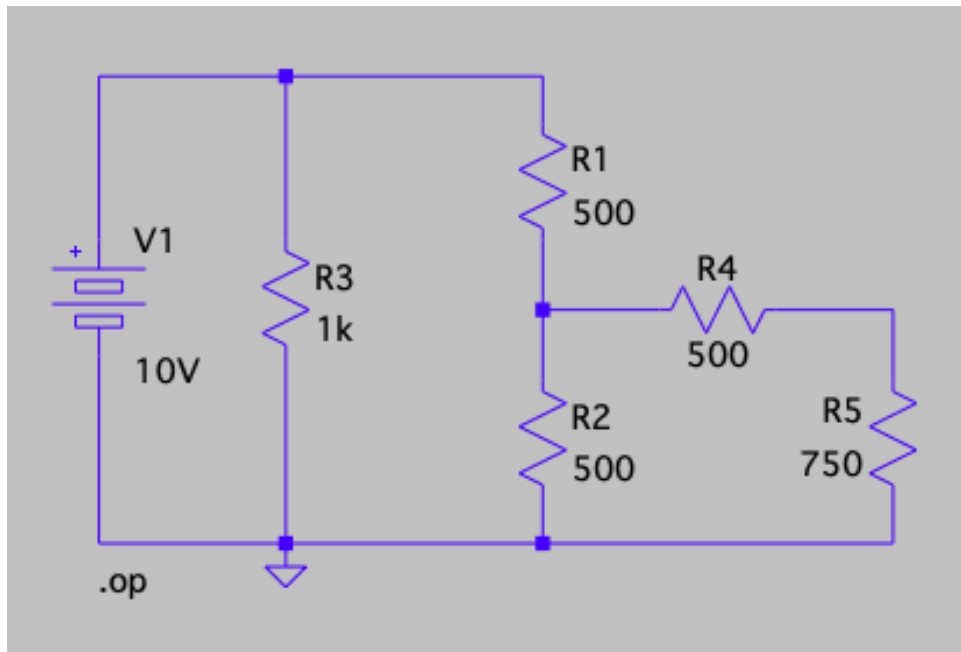
- Operating point simulations calculate the DC voltages at all nodes of the circuit.
- Time domain transient analysis determines conditions for a set time, determined by the user, e.g. the input and output voltages of an AC circuit.
- Frequency domain AC analysis allows the investigation of a circuit over a range of source frequencies, e.g. 10 Hz to 1 MHz. The resulting graphs may be linear or logarithmic plots of the circuit transfer function.

Note that the theoretical formulae given in parts 2, 3 are covered in the Week 2A course notes.

Part 1. DC Circuit Analysis

Open LTspice, start a new schematic, and draw and save the following schematic. To begin, make the load resistor (R5) 10 M Ω instead of 750 Ω (use 10e6 or 10Meg for the value), to replicate an open circuit at the load point. This is necessary, since the simulation will not accept open loops or nets.

Many common components, such as resistors and the ground symbol, are available in the top menu bar, but you must use the generic “Component” button to place a voltage source (called either “voltage” or “battery”). You can use control-R to rotate components before placing them. After placing a component, right click on it to change its parameters. **Don’t forget the circuit ground.**



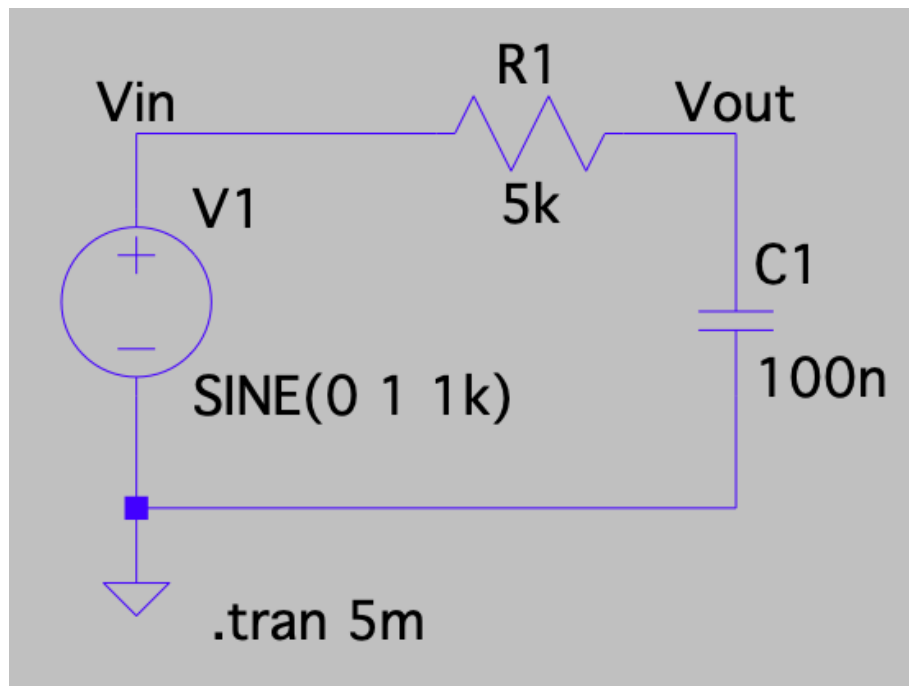
Run the simulation by clicking . A window will appear. From there, go to the operating point simulation on the right-most tab and press accept. Right away, another window will appear with text indicating all the currents and voltages of the system. **However, you will note that it is very hard to identify which voltages correspond to which nodes. To fix this, add a label called Vout in the node in between R4 and R5 (use “Label Net” in the top bar beside the ground button).** Re-run the simulation to see the voltage at the node labelled Vout.

This gives the Thevenin voltage, V_{Th} , across the 10 M Ω (effectively an open-circuit output).

Modify the circuit to replace the load with its original value (750 Ω), and repeat, again recording the voltage and current levels.

Task: using this simulation, determine the Thevenin resistance, R_{Th} , for the circuit, knowing the load voltage and V_{Th} . Explain the process of calculating the Thevenin resistance and voltage using the simulation.

Part 2. Time Domain (Transient) Analysis



Start a new circuit and on the schematic page draw the circuit shown above. Components are as follows:

R1 = 5 k Ω (value 5000 or 5k);

C1 = 100 nF or 0.1 μ F (value 100n or 0.1u);

V1 = select the “voltage” component, right-click it and select “Advanced options”, then use the SINE function. Set DC offset to 0, amplitude to 1 and frequency to 1k.

Now, run the simulation, selecting a transient simulation with a 5ms stop time. A plotting window will appear. Click on the circuit to probe Vout and Vin: two sine waves should appear. If successful, note the decaying transient in the output voltage; the peak positive amplitude starts around 0.42 V and decays to about 0.3 V by 2.5 ms. (Click on the coloured title for the trace to bring up cursors to measure the voltage of that trace).

Since the transient decays to zero in about 2.5 ms, zoom in on the area from 3 ms to 5 ms to display the traces in their “steady state” condition. Alternatively, in the simulation window, edit the simulation profile (right click the “.tran” SPICE directive text) to “start saving data after” 3 ms, and run the simulation again.

To calculate the phase difference, ϕ , between the input and output, first use the cursors to measure the difference between the times where each trace crosses 0V, Δt (ensure you are using like points—either both crossing upwards or both crossing downwards). Then measure the period of the input trace (T). Calculate the phase difference between the traces from these values as follows:

$$\phi = \frac{\Delta t}{T} 360^\circ$$

Record the measured phase difference, and the theoretical phase difference, where the latter is calculated from:

$$\phi_{\text{th}} = -\arctan(\omega RC), \omega = 2\pi f, f = 1000\text{Hz}$$

Measure also the gain of the circuit, $A_v = V_{\text{out}}/V_{\text{in}}$, i.e. ratio of the peak values, and calculate the theoretical gain, determined from:

$$A_v = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

Task: show the process of calculating the phase and gain of the circuit, both theoretically and using the simulation results.

Part 3. AC Sweep Analysis

Set up a new circuit and redraw the circuit from part 2. However, this time, we will change the simulation type of to an AC analysis. Before changing the simulation type, right click on the voltage source and on the new window that appears locate the “Small Signal parameters” and set “AC Amplitude” to 1.

Now try to run the simulation and set it to an AC analysis. The start frequency should be 10Hz and the end frequency to 10MHz, using 5 points per decade.

The resulting graphs are: (a) the gain and (b) the phase-angle of the output with respect to the input – both on the same graph. The gain will be in Decibels (dB), and the phase angle will be in degrees.

Run the simulation and do the following measurements using the cursors:

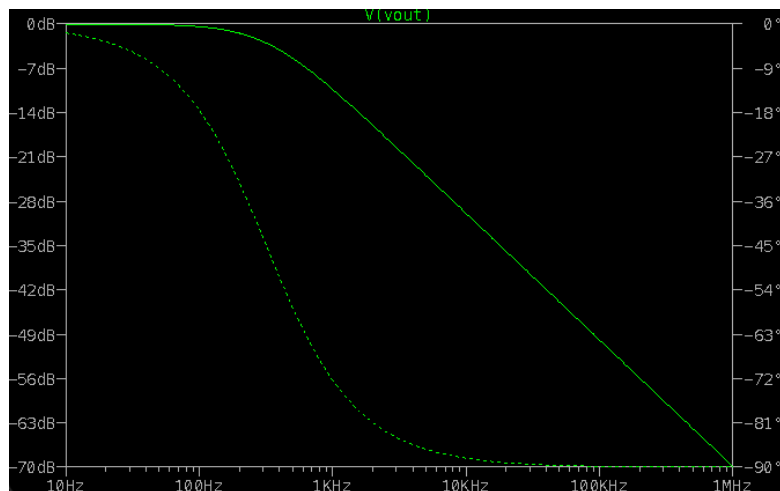
- Cutoff Frequency (f_c) – find the frequency where the Gain is -3 dB
- Phase angle (ϕ) at this frequency f_c

Compare with the theoretical values:

$$f_c = \frac{1}{2\pi RC}$$

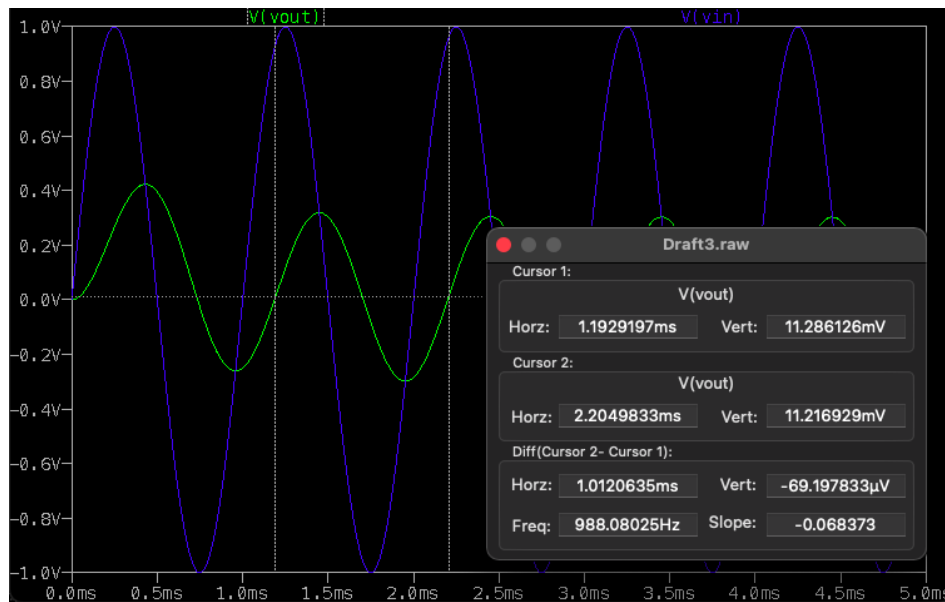
$$\phi = -\arctan(2\pi f_c RC)$$

The screen should now look like:



Task: using the simulated graphs, show the process of calculating the simulated and theoretical cutoff frequency and phase

Appendix 1 – Use of the Probe Cursors



In LTspice there are two cursors that can be placed in any of the drawn graphs. To place the first cursor, left click the graph label on the top on the desired graph. A second cursor can be placed in the same way using a double-click instead of one.

The cursors can be moved around by clicking and dragging on the crosshairs. The cursor number appears if you are hovering in the correct part of the cursor.

The measured values appear in the pop up window as shown above.